# PAILA AKASH

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## PROFESSIONAL SUMMARY

Third-year Computer Science undergraduate (CGPA 9.45/10) specializing in generative AI, full-stack web development, and RF signal intelligence. Currently interning at DRDO's Defence Research & Development Laboratory (DRDL), with counter-UAV detection experience at the Indian Army's 509 Army Base Workshop. Co-author of an IEEE conference submission on cognitive digital twins, with experience in open-source software and applied machine learning.

## EDUCATION

**K L University, Hyderabad** — Hyderabad, India
*Bachelor of Technology, Computer Science and Engineering (Gen AI)* — **CGPA 9.45 / 10**

## EXPERIENCE

**AI/ML Intern** — Hyderabad, India
*Defence Research & Development Laboratory (DRDL), DRDO* — Nov 2, 2026 - Present (expected end Feb 2, 2027)

- On-site internship under the guidance of Ranajoy Pal, Scientist 'F', DWST.

**Intern - 509 Army Base Workshop, Indian Army (Ministry of Defence)** — Agra, India
*Counter-UAV RF Detection & Direction-Finding Project* — Jun 2026 - Present

- Contributing to an RF-based UAV detection, direction-finding, and distance-estimation system built around a KrakenSDR coherent receiver, a four-element antenna array, and a Raspberry Pi 5 (8GB), targeting drone signals in the 2.4 GHz and 5.8 GHz bands.
- Implemented MUSIC-based Direction-of-Arrival (DoA) estimation and RSSI-based signal-strength analysis to compute azimuth angle, elevation angle, bearing direction, and approximate target distance.
- Used MATLAB and Simulink for signal processing, mathematical modeling, and simulation, and built a real-time monitoring dashboard with LED/buzzer-based threat alerting.
- Designed and trained an RF signal-classification model that ingests uploaded RF recordings and predicts signal category (Wi-Fi, Bluetooth, LTE, or Drone RF) with a confidence score, alongside live RSSI, peak power, and noise-floor readouts.

## PROJECTS

**Counter Drone RF Analyzer / Drone DF Simulator** — GitHub

- Desktop application to upload and analyze RF recordings, visualize live power spectrum and spectrogram, and output AI-based drone-presence prediction with confidence, RSSI, peak power, and noise floor.

**LilNest - Maternal Care Platform** — Live Demo

- Built a full-stack maternal-health web application using React, Node.js, and MongoDB with authentication, responsive UI, and RESTful API integration between frontend and backend.

**MotionBeatz - Gesture-Controlled Music Player**

- Developed a computer-vision-based music player that uses hand-gesture recognition to control playback, volume, and track navigation.

## PUBLICATIONS

- Co-author, *"NeuroCloak: Cognitive Digital Twin (CDT) for AI Systems,"* 2026 IEEE 8th International Conference on Cybernetics, Cognition and Machine Learning Applications (ICCCMLA) - currently under review.
- Proposed a Cognitive Digital Twin framework combining neuro-symbolic AI and metacognitive monitoring to make AI decision-making self-evaluating, bias-auditable, and transparent for high-stakes domains such as healthcare and defense.

## OPEN SOURCE CONTRIBUTIONS

- **freeCodeCamp** - Contributed a verified, merged commit adding interactive examples to the functional-programming curriculum lesson (PR #63872).
- Active contributor to community-driven open-source repositories on GitHub, focused on improving real-world code quality and documentation.

## CERTIFICATIONS

- **Automation Anywhere** - Certified Advanced Automation Professional
  Issued Aug 8, 2026 | Expires Aug 8, 2028 | Credential ID AAADVC2024A360 - 190987243
- **MongoDB, Inc.** - MongoDB Certified Associate Developer
  Issued Nov 21, 2025 | Credential ID MDBhtilqokyju
- NVIDIA - Certificate of Competency, "Accelerating End-to-End Data Science Workflows"

## ACHIEVEMENTS

- Selected participant, IIT Madras Workshop
- Honors Roll Student - recognized for sustained academic excellence

## TECHNICAL SKILLS

- **Languages** Java, C++, Python, JavaScript, C
- **Frameworks** React.js, Node.js, Express.js, FastAPI, Tailwind CSS
- **Databases** MongoDB, PostgreSQL, MySQL
- **Developer Tools** Git, GitHub, VS Code, Postman, Render
- **AI / ML** Data Preprocessing & Feature Engineering, Computer Vision, Machine Learning Algorithms
- **Core CS** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
- **Soft Skills** Problem-Solving, Team Collaboration, Communication, Project Management, Critical Thinking, Adaptability